

AP CSP | Lesson 20: Searching Algorithms

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Q5 requires exactly TWO choices. Trace statements in their execution order. Each program statement and each brace occupies its own line. Show reasoning and use the stated valid input range.

1. For values ← [3,8,12], what is values[1]? [1 mark]

A. 3 B. 8 C. 12 D. 1

Answer: _____

2. Which property allows an ascending binary search to discard all values left of a midpoint when the target is greater than that midpoint? [1 mark]

A. All values are positive. B. The list is sorted in ascending order.
C. The list length is odd. D. The target must be present.

Answer: _____

3. An early-stopping linear search checks [3,8,12,19,27,34,50] from the beginning for target 34. How many elements are checked? [1 mark]

A. 2 B. 5 C. 6 D. 7

Answer: _____

4. Binary search checks the middle value 19 in [3,8,12,19,27,34,50] for target 34. Which candidates remain? [1 mark]

A. 3,8,12 B. 19,27,34,50 C. 34 only D. 27,34,50

Answer: _____

5. Which TWO statements are correct? Select exactly TWO. [1 mark]

A. Linear search needs sorted data. B. Binary search can conclude that a target is absent.
C. Different correct searches can return the same membership result with different costs. D. Binary search always uses fewer checks than linear search for every target.

Answer: _____

6. A sorted list has 31 distinct elements. Binary search checks one midpoint at each step. The possible interval sizes before checks are 31,15,7,3,1. What is the maximum number of checks? [1 mark]

A. 4 B. 5 C. 15 D. 31

Answer: _____

AP CSP | Lesson 20: Searching Algorithms

Written responses | Questions 7-9

7. [4 marks]

Search [3,8,12,19,27,34,50] for 50 using binary search. Give each inspected value in order and the total number of element checks.

8. [4 marks]

Search [3,8,12,19,27,34,50] for 28 using binary search. Give the three inspected values and explain why the final result is false.

9. [4 marks]

An incorrect search uses [9,1,7,3,5] for target 9. It checks the middle value 7, then keeps only the values to its right because $9 > 7$. Identify the broken prerequisite, the remaining values, the lost target, and one valid repair.

AP CSP | Lesson 20: Searching Algorithms

Written responses | Questions 10-12

10. [4 marks]

For [3,8,12,19,27,34,50], compare early-stopping linear and binary search for target 3. State each count, whether the membership results agree, and why this disproves a claim that binary is faster on every individual input.

11. [4 marks]

Two algorithms answer whether a target is in a list. State when early-stopping linear search returns true, when it returns false, when binary search returns false, and what it means for their results to agree for the same valid input.

12. [4 marks]

A student performs one search on an unsorted list today. Another application performs many searches on an already sorted list. Recommend a suitable method for each, give one reason for each recommendation, and address preparation cost.
